

Astronomy 102 - Lab 9

Light & the Doppler Effect

Matthew Prest



Any guesses when this was taken?



What do you notice?



ARTEMIS II

First Crewed Test Flight to the Moon Since Apollo

- 1 LAUNCH**
Astronauts lift off from Launch Pad 39B at Kennedy Space Center.
- 2 ORION UPPER STAGE SEPARATION (USS) BURN**
Begins high Earth orbit checkout. Life support, exercise, and habitation equipment evaluations.
- 3 PERIGEE RAISE BURN**
- 4 PERIGEE RAISE MANEUVER**
- 5 APOGEE RAISE BURN TO HIGH EARTH ORBIT**
Begin 23.5-hour checkout of spacecraft.
- 6 ORION SEPARATION FROM INTERIM CRYOGENIC PROPULSION STAGE (ICPS) FOLLOWED BY PROX OPS DEMO**
Plus manual handling qualities assessment for up to 2 hours.
- 7 JETTISON SOLID ROCKET BOOSTERS, FAIRINGS, AND LAUNCH ABORT SYSTEM**
- 8 CORE STAGE MAIN ENGINE CUT OFF**
With separation.
- 9 OUTBOUND TRANSIT TO MOON**
Outbound trajectory correction (OTC) burns as necessary for lunar free return trajectory; travel time approximately 4 days.
- 10 TRANS-EARTH RETURN**
Return trajectory correction (RTC) burns as necessary to aim for Earth's atmosphere; travel time approximately 4 days.
- 11 CREW MODULE SEPARATION FROM SERVICE MODULE**
- 12 ENTRY INTERFACE (EI)**
Enter Earth's atmosphere.
- 13 SPLASHDOWN**
Ship recovers astronauts and capsule.
- 14 LUNAR FLYBY**
4,047 mi/6,513 km (mean) lunar far side flyby altitude.
- 15 ICPS EARTH DISPOSAL**
ICPS deploys CubeSats.
- 16 ICPS EARTH DISPOSAL**
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- 17 ICPS EARTH DISPOSAL**
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PROXIMITY OPERATIONS DEMONSTRATION SEQUENCE	
1	9
2	10
3	11
4	12
5	13
6	14
7	15
8	16
17	

The latest in Astronomy

Artemis II mission!

- First crewed beyond low Earth orbit since 1972
- Scientific testing and instrument evaluation of new space technology
 - Biological sample testing
 - Laser communication
- Set the record for human distance from Earth, 252,756 miles (406,771 km)
- Future Artemis missions to include lunar surface landings and preparation for extended human presence



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- Wake up song tradition, chosen by friends and family of the crew members

Flight day ↕	Song ↕	Artist ↕
Day 1	"Sleepyhead" (Passion Pit Cover)	Young & Sick
Day 2	"Green Light"	John Legend, André 3000
Day 3	"In A Daydream"	Freddy Jones Band
Day 4	"Pink Pony Club"	Chappell Roan
Day 5	"Working Class Heroes (Work)"	CeeLo Green
Day 6	"Good Morning"	Mandisa, TobyMac
Day 7	"Tokyo Drifting"	Denzel Curry, Glass Animals
Day 8	"Under Pressure"	Queen, David Bowie
Day 9	"Lonesome Drifter"	Charley Crockett
Day 10	"Run to the Water"	Live
	"Free"	Zac Brown Band

Sound & the Doppler Effect

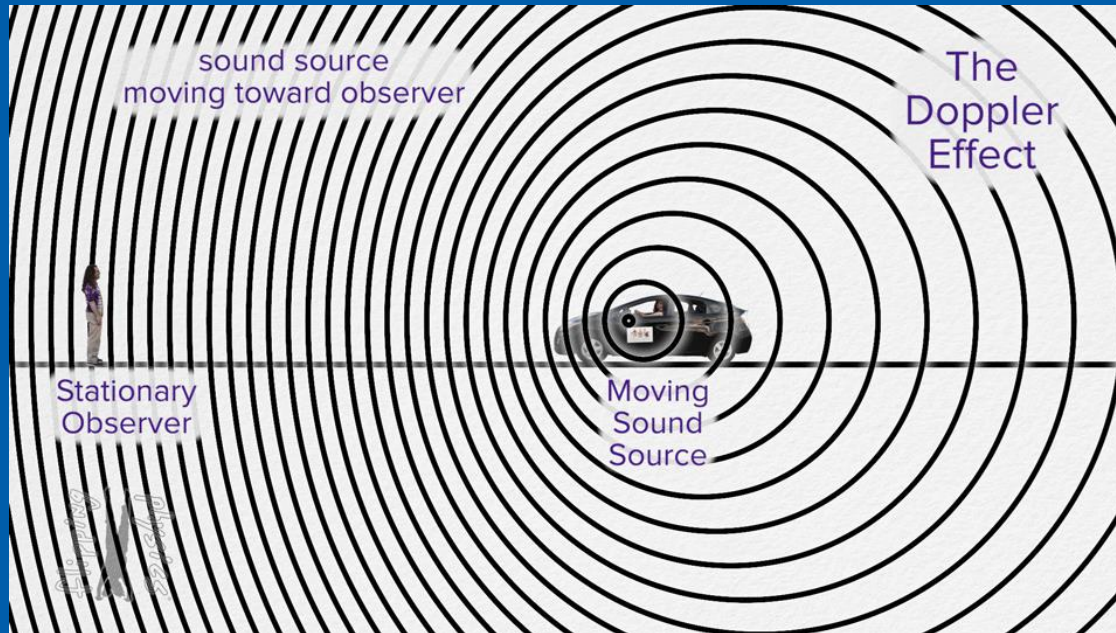
Why does a motorcycle sound different when it is coming towards you vs driving away from you?



Sound & the Doppler Effect

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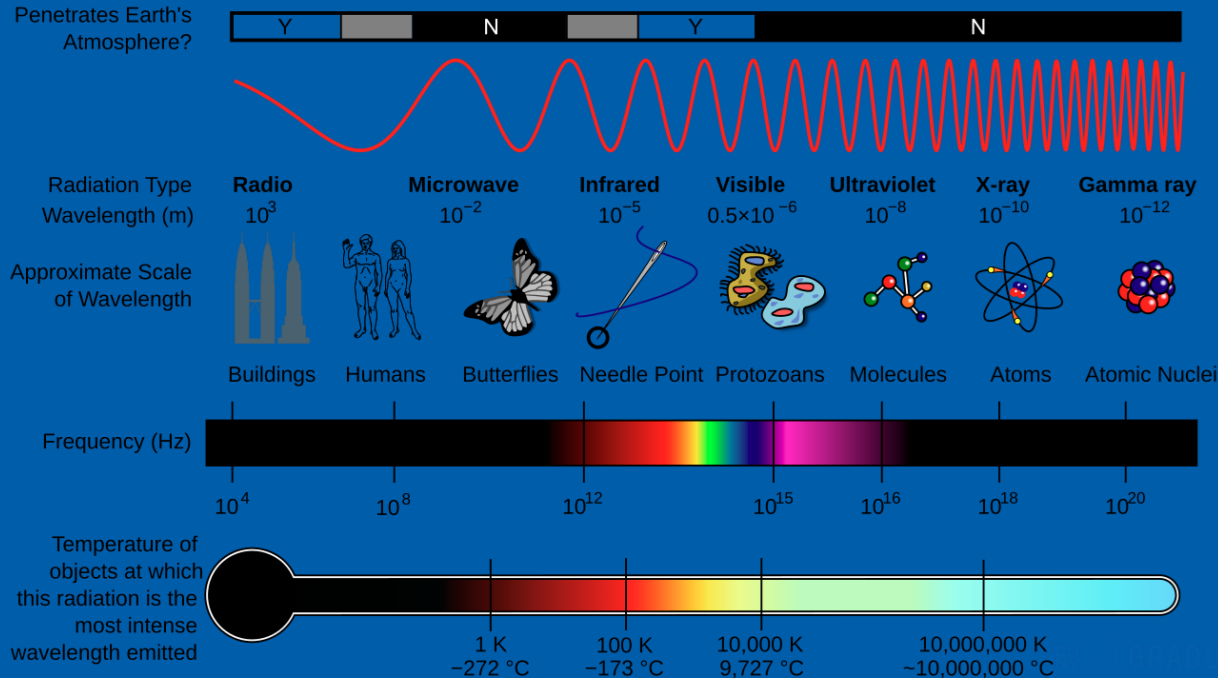
- The Doppler Effect!
- Sound waves in front are compressed by the moving object, while sound waves behind are stretched. This changes the frequency of the wave which we hear as pitch.



Doppler Effect & Light

The same thing happens with any wave, including light!

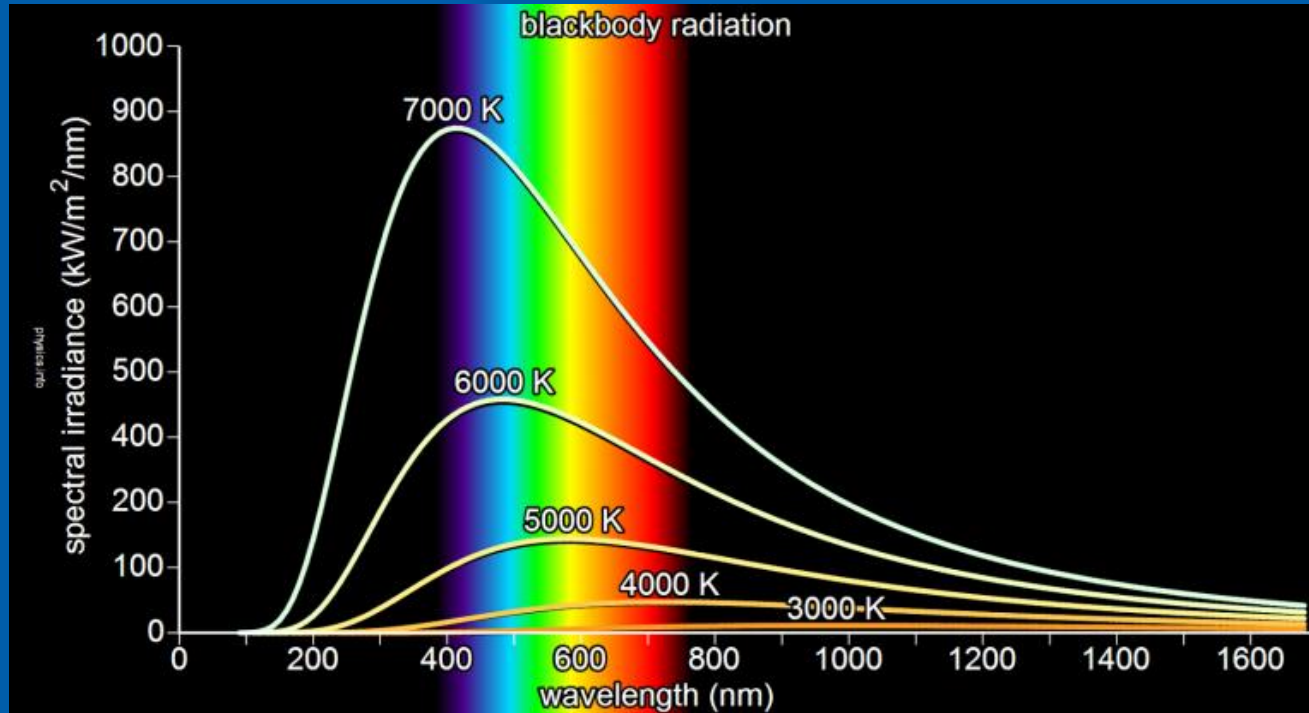
- For visible light this causes colour change aka “Red Shift”



Doppler Effect & Light

This can affect thermal radiation and the perceived temperature!

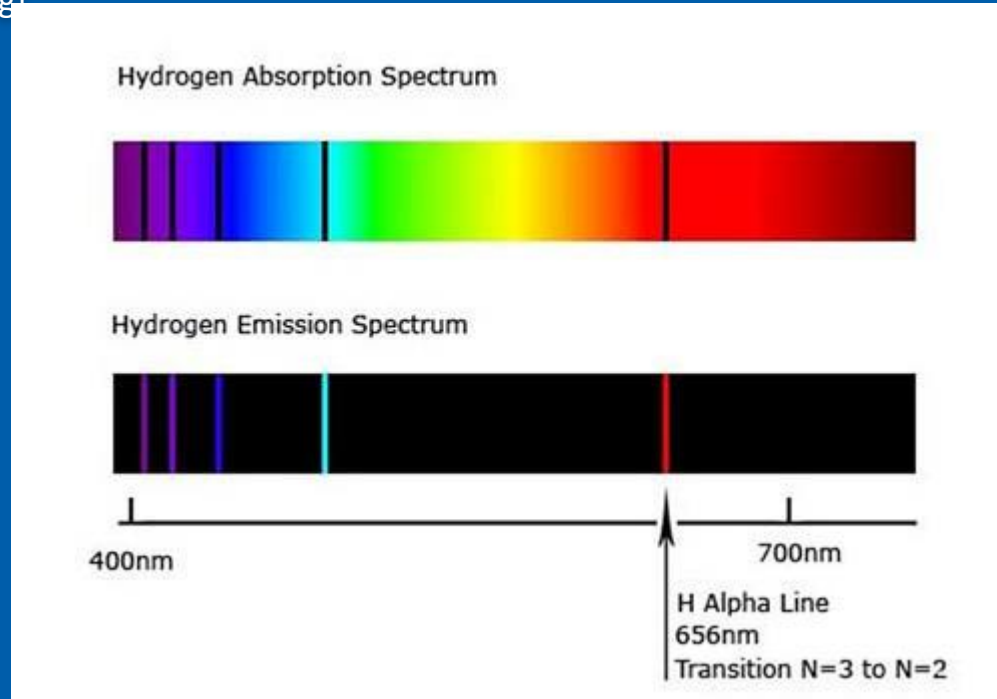
- Remember why hot things glow?



Doppler Effect & Light

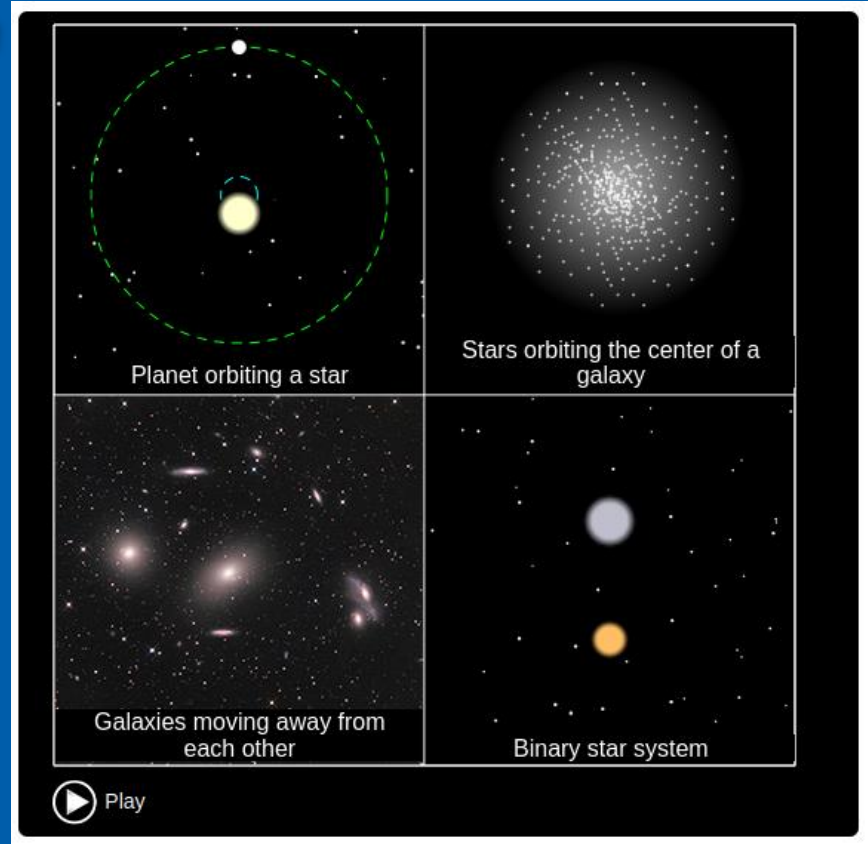
This can affect Emission vs Absorption Spectra

- If we know the substance, and we measure the spectrum we can learn how fast it is moving!



Pearson Self-guided Tutorial

Launch the self-guided tutorial and follow along!



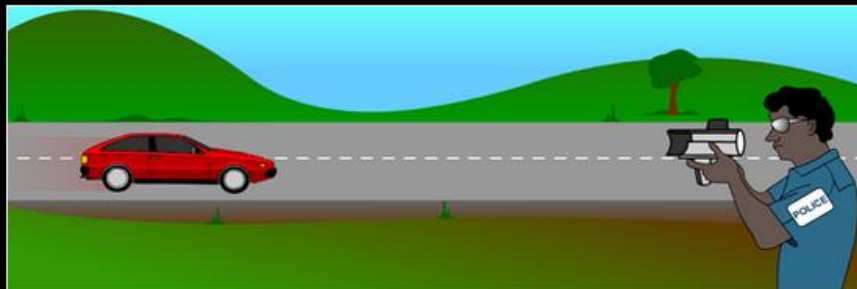
Pearson Self-guided Tutorial

Make a Prediction

Feel free to change your prediction as you progress through this tutorial. To do this, select the prediction link in the top right corner of the screen.

How do you think a traditional radar gun is able to determine the speed of a moving car?

- ☐ The speed of the car is determined from how much the speed of the returning radar beam changes.
- ☐ The speed of the car is determined from how much the wavelength of the returning radar beam changes.
- ☐ The speed of the car is determined from how much the intensity of the returning radar beam changes.
- ☐ A radar gun can't be used to determine the speed of the car, only the distance from the car to the radar gun.

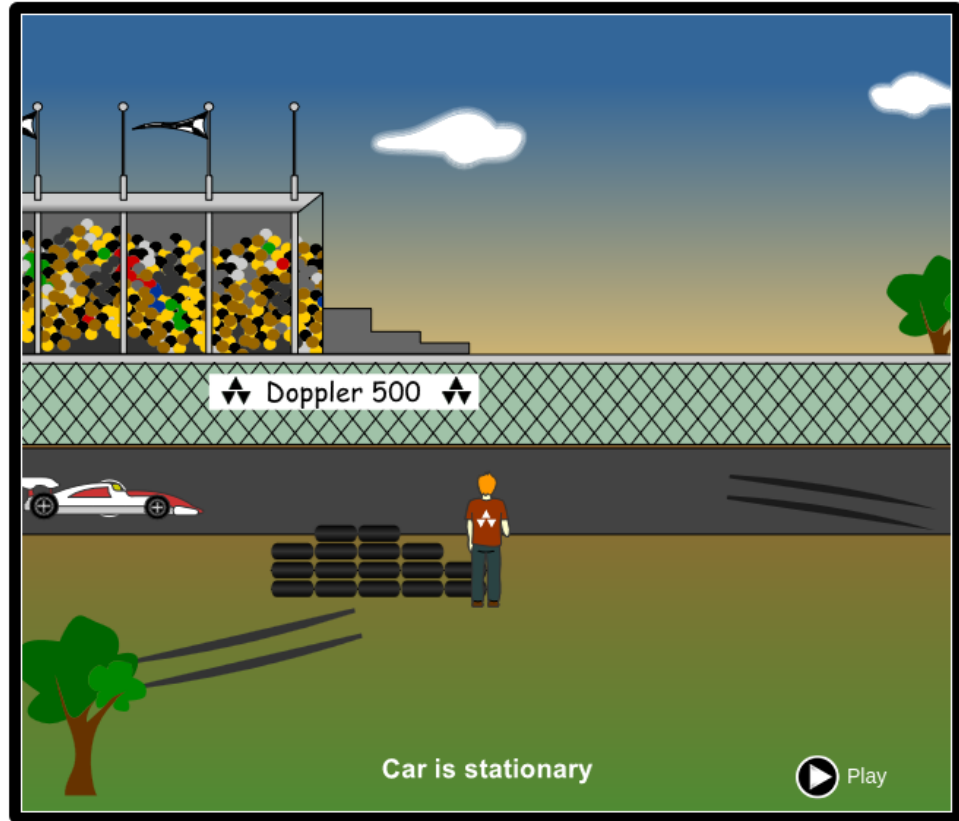


Pearson Self-guided Tutorial

Pitch and Frequency

The pitch of a sound depends solely on the sound's frequency: the higher the frequency, the higher the pitch.

This means that the observed frequency of the sound wave emitted by the car is higher when the car is traveling toward you, and lower when the car is moving away from you.

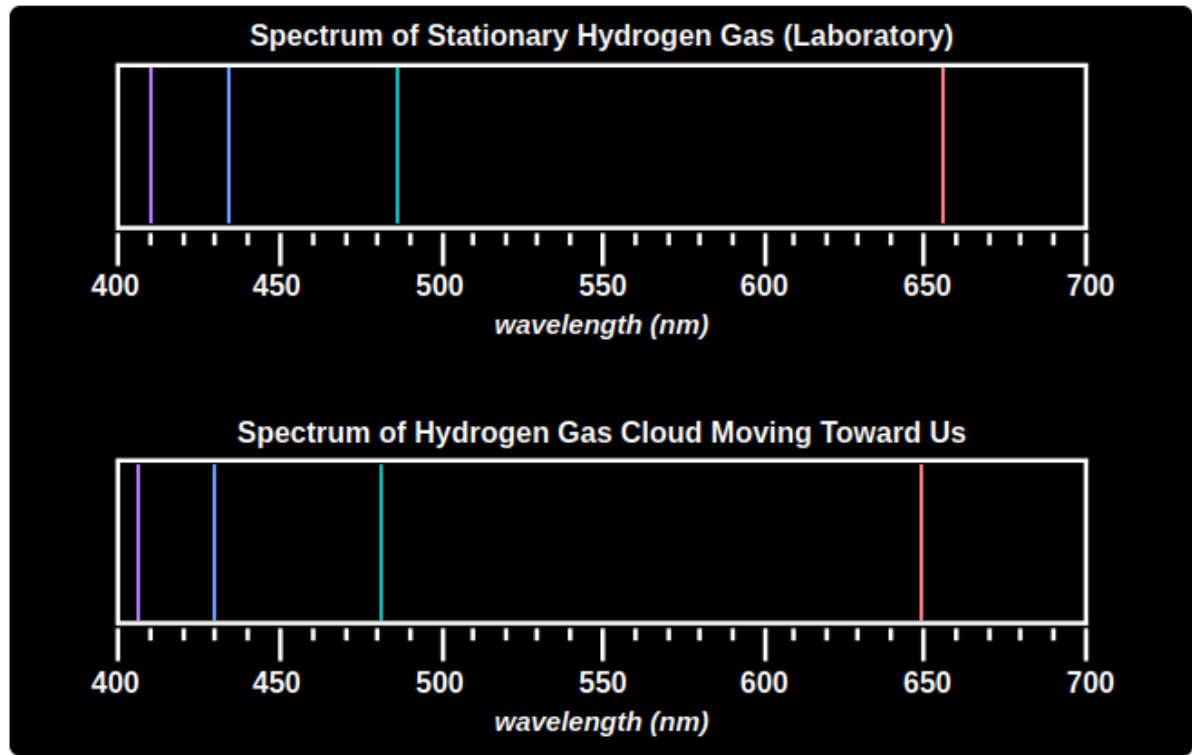


Pearson Self-guided Tutorial

Spectrum of Stationary Hydrogen

In the previous example, the source of light (the laser) emitted radiation having only one wavelength. If the source is emitting light with more than one wavelength, each wavelength will be Doppler shifted.

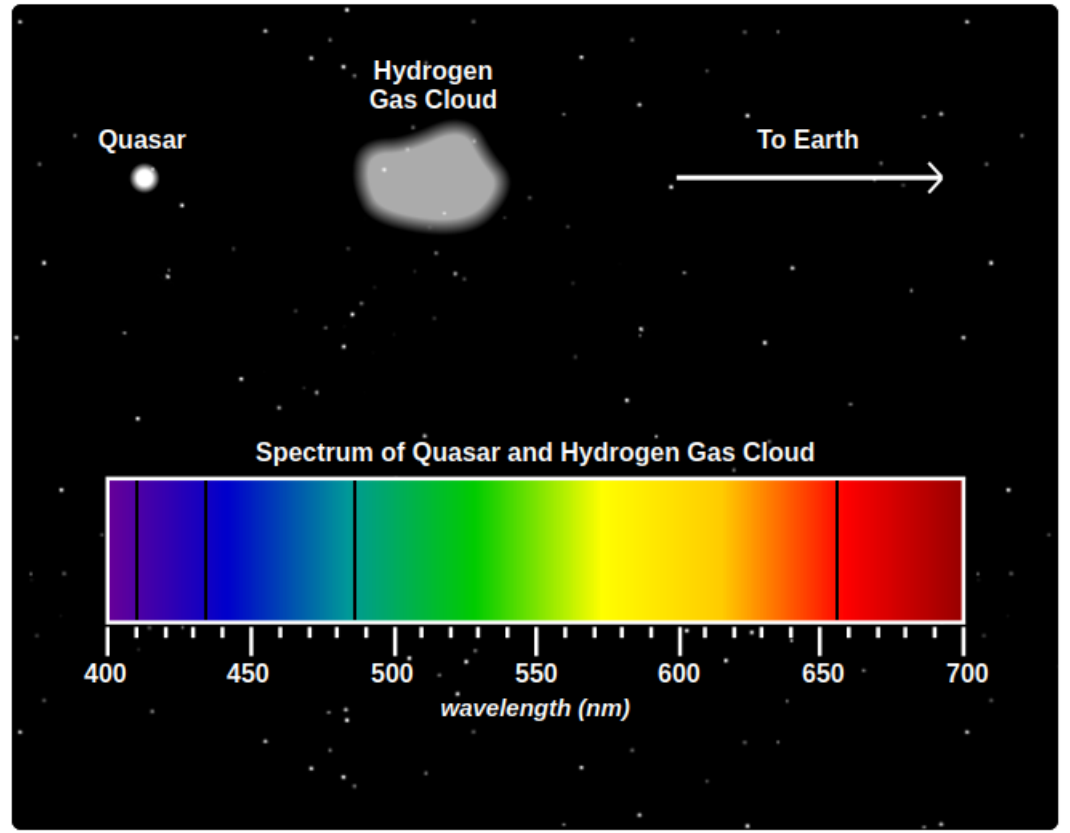
For example, the spectrum of a hot hydrogen gas contains several emission lines. If an interstellar cloud of hot hydrogen gas is moving toward us, the wavelength of each emission line is blueshifted (each wavelength will be smaller than the equivalent line from a stationary gas cloud of hydrogen).



Pearson Self-guided Tutorial

Absorption Lines

Absorption lines can also be used to determine the speed of an object. For example, if a cold intergalactic hydrogen cloud is between us and a bright source emitting a continuous spectrum, we will see absorption lines. These lines will be Doppler shifted if the cloud is moving with respect to us.



Reminders

1. Complete the Tutorial first!

- This is best done now in class where I am available to assist

1. Attempt the essay questions last!

- These should be based on what you have covered in the tutorial.

1. Previous week's lab report due at the beginning of the next lab!

- Final grades are based on best 6 online labs plus each of the 4 in-class labs

1. Grades will be posted to Brightspace

- Check now that you have access to our course!
- Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!

1. Office hours

- Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!